

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Fifth Semester of B.Pharm. Examination
University Theory Examination April 2016
MA331 Bio-Statistics

Date: 13/04/2016 Time: 10:00 a.m. to 01:00 p.m. Maximum Marks: 80
Wednesday

Instructions:

- (1) Figures to right indicate marks.
- (2) Section wise separate answer book to be used.
- (3) Draw neat sketches wherever necessary.
- (4) Use of scientific non-programable calculator is allowed.

SECTION I

Q.1. Attempt any TWO of the following.

(a) Answer the following questions. [10]

- 1) Define the term Biostatistics and write its importance with examples in pharmacy. (05)
- 2) If $x_1, x_2, \dots, x_n$ is a sample of n values of variable x , Then match each measure with correct formula. (05)

Measure	Formula	
(a) Arithmetic Mean	(1) $(\prod_{i=1}^n x_i)^{1/n}$	(4) $\sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}}$
(b) Geometric Mean	(2) $\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$	(5) $\frac{\sum_{i=1}^n x_i}{n}$
(c) Standard deviation	(3) $\frac{\sum_{i=1}^n x_i - \bar{x} }{n-1}$	
(d) Mean deviation		

(b) Write with explanations whether following statements are true or false. [10]

- 1) The standard deviation for two series 1, 4, 7 and 2, 8, 14 are identical.
- 2) If $\sum_{i=1}^4 (x_i - 5) = 0$ then $\bar{x} = 5$
- 3) The geometric mean of 2, 4, 8, 16, 32 is 8.
- 4) The mode of 5, 5, 5, 4, 10 is 10
- 5) The arithmetic mean of 4, 4, 4, 4 is 4

(c) Answer the following questions. [10]

1) Write short note on Population data and Sample data in Biostatistics. (05)

2) Write important properties of arithmetic mean. (05)

Q.2. Attempt any TWO of the following.

(a) The data in Table (1) represent frequency distribution of alkaline phosphatase level of the subjects 65 years of age and older. Construct using frequency table (1) [10]

Table 1: Frequency Distribution:alkaline phosphatase

Lower Limit	40	60	80	100	120	140	160	180	200
Upper Limit	59	79	99	119	139	159	179	199	219
Frequency	16	51	52	31	16	6	3	1	2

1) Relative Frequency distribution (03)

2) Cumulative frequency distribution and (03)

3) Histogram (04)

(b) Quinolones are antibacterial drugs widely used to combat various important diseases of farmed fish. Catfish muscle was fortified with each quinolone at ppb levels. The recovery results for one of two of the additives (at approximately 20 ngg^{-1}) are shown in Table (2). Compute [10]

Table 2: Residue recovery rates (%)

Oxolinic acid	95.4	103.0	97.0	94.9	96.5	95.9
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1) Arithmetic mean and standard deviation % recovery. (05)

2) Mean square deviation about 100.0 % recovery . (05)

(c) The following figures (Table 3) refer to the concentration of albumin, in g l^{-1} , in the blood sera of 8 healthy adults: Compute [10]

Table 3: Concentration of albumin, in g l^{-1}

37	39	37	42	39	45	42	39
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1) All quartiles albumin concentration (05)

2) Mean deviation about median albumin concentration. (05)

SECTION II

Q.3. Attempt any TWO of the following.

- (a) A large number of determinations was carried out on the same sample and the results are known to be normally distributed with $\mu = 215$ and $\sigma = 35$. What percentage of determinations will fall [10]
 1) between the boundaries 200 and 250? 2) below 200? 3) above 260?
 You are given $P(Z < -0.43) = 0.3336$, $P(Z < 1) = 0.8413$, $P(Z < 1.29) = 0.9015$.
- (b) Assume the number of episodes per year X of otitis media, a common disease of the middle ear in early childhood, follows a Poisson distribution with parameter $\lambda = 1.6$ episodes per year. Compute the probability that number of episodes of otitis media in the first year of life is [10]
 1) zero 2) less than two 3) more than or equal to two
- (c) Write the assumptions of Binomial distribution. Compute the probability of k lymphocytes out of 5 white blood cells if the probability that any one cell is a lymphocyte is 0.2, $k = 0, 1, 2, 3, 4, 5$ [10]

Q.4. Attempt any TWO of the following.

- (a) In a stability testing programme samples of a drug product were stored at 25°C and 60% relative humidity. Determinations of the potency of the drug product were made at seven time points: zero, 3, 6, 9, 12, 18 and 24 months after manufacture. The reported results are percentage of label claim shown in Table (4) [10]

Table 4: The stability data

Month	0	3	6	9	12	18	24
Potency (%)	102.6	100.4	98.4	99.4	99.8	97.8	97.6

- 1) Construct the scatter diagram from the data and write interpretation.
- 2) Obtain regression line to predict average % potency of drug product two years after manufacture.
- 3) Determine the time when potency of drug product will have fallen to 95% of its label claim using regression line obtained in question (2)

(b) Seven measurements of the pH of a buffer solution gave the following results:

Table 5: pH measurements

5.12	5.20	5.15	5.17	5.16	5.19	5.15
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If μ denotes the true mean pH then determine 95% and 99% confidence interval for μ . You are given $t_{6,0.05/2} = 2.4469$, $t_{6,0.05} = 1.9432$, $t_{6,0.01/2} = 3.7074$, $t_{6,0.01} = 3.1427$

[10]

(c) Consider, the data from biomedical analysis in Table (6). They represent Cu , Mn and Zn concentrations determined in 8 different structures of the human brain. [10]

Table 6: Concentration of Cu , Zn and Mn in different brain structures

Brain Structure	1	2	3	4	5	6	7	8
Cu	25.8	24.2	27.3	32.8	27.3	17.9	14.0	13.3
Mn	1.0	1.0	1.1	1.5	1.8	1.2	1.1	1.0
Zn	78.0	81.8	69.4	76.1	62.5	60.1	34.2	35.5

- 1) Construct scatter diagram of variables Cu and Zn and write the interpretation.
- 2) Compute simple correlation between the variables Cu and Zn